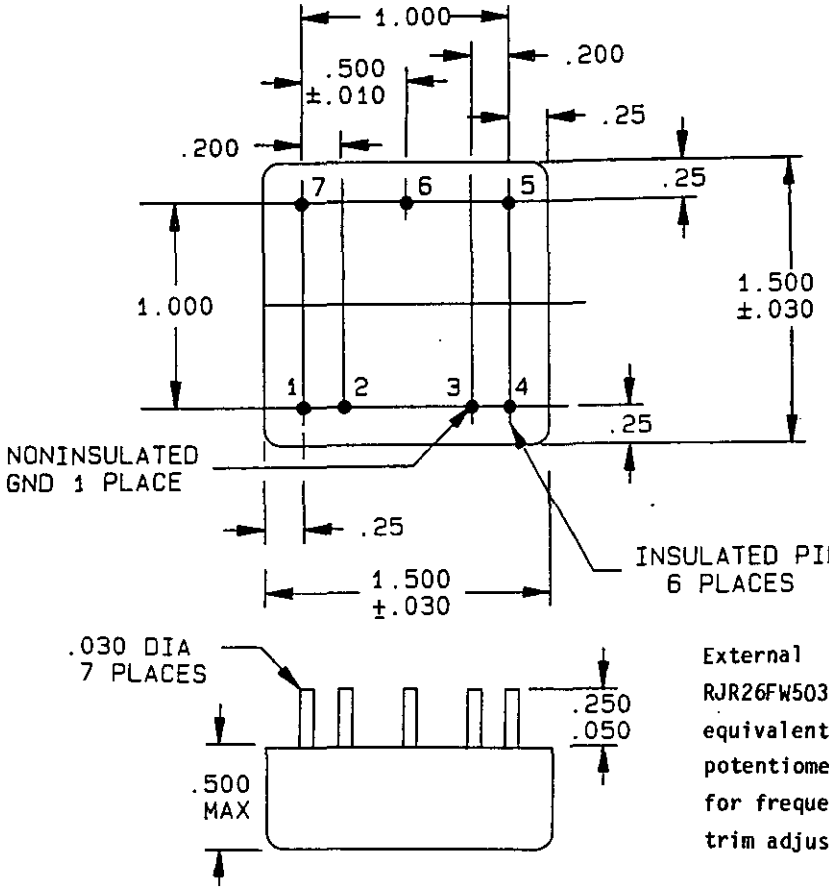
MILITARY SPECIFICATION SHEET

OSCILLATORS, CRYSTAL CONTROLLED, TYPE 3, (TEMPERATURE COMPENSATED (TCXO)), 10.0 MHz HERMETIC SEAL, SQUAREWAVE, CMOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirment for acquiring the oscillators described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-O-55310.



Inches	mm
.010	0.25
.020	0.51
.030	0.76
.050	1.27
.200	5.08
.245	6.22
.250	6.35
.500	12.70
1.000	25.40
1.500	38.10

Pin	Function
1	Osc. supply
2	Logic supply
3	GND
4	Output
6	Trim
5	Trim
7	Trim

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Design and construction.

Package: See figure 1.

Mounting: See figure 1.

Terminals: See figure 1.

Weight: 2.0 ounces (57 grams-mass), maximum.

Construction technology: Discrete components.

Quartz crystal: AT-cut, swept cultered quartz.

Overall frequency accuracy: ±2 ppm/year.

Calibration: Manufacturer/user frequency offset calibrated.

Temperature:

Operating: -55°C to +85°C
Storage: -62°C to +125°C

Load test circuit: See figure 2.

Design for nuclear survivability: Design to withstand the following levels:

Total ionizing dose: 10^3 to 10^4 rads (Si)
Peak prompt gamma flux: 1×10^5 to 3×10^{10} rads (Si)/s
Neutron fluence: 5×10^{11} to 9×10^{12} neutrons/cm^3

Seal: Hermetic, in accordance with MIL-O-55310.

Nominal frequency: 10 MHz.

Oscillator supply voltage: 10 V dc ±3%, peak to peak ripple of 50 mV at a ripple frequency of 1 Hz to 1 MHz.

Logic supply voltage: 5.0 V dc ±10%.

Overvoltage survivability: In accordance with MIL-O-55310.

Oscillator input current: 1.5 mA maximum at 10.3 V dc. Measurement taken at -55°C, +23°C and +85°C, and nominal load.

Logic input current: 1.5 mA maximum at 5.5 V dc (with load, see figure 2).

Frequency adjustment: External using a RJR26FW503P potentiometer, range ±3.0 ppm, resolution ±0.1 Hz.

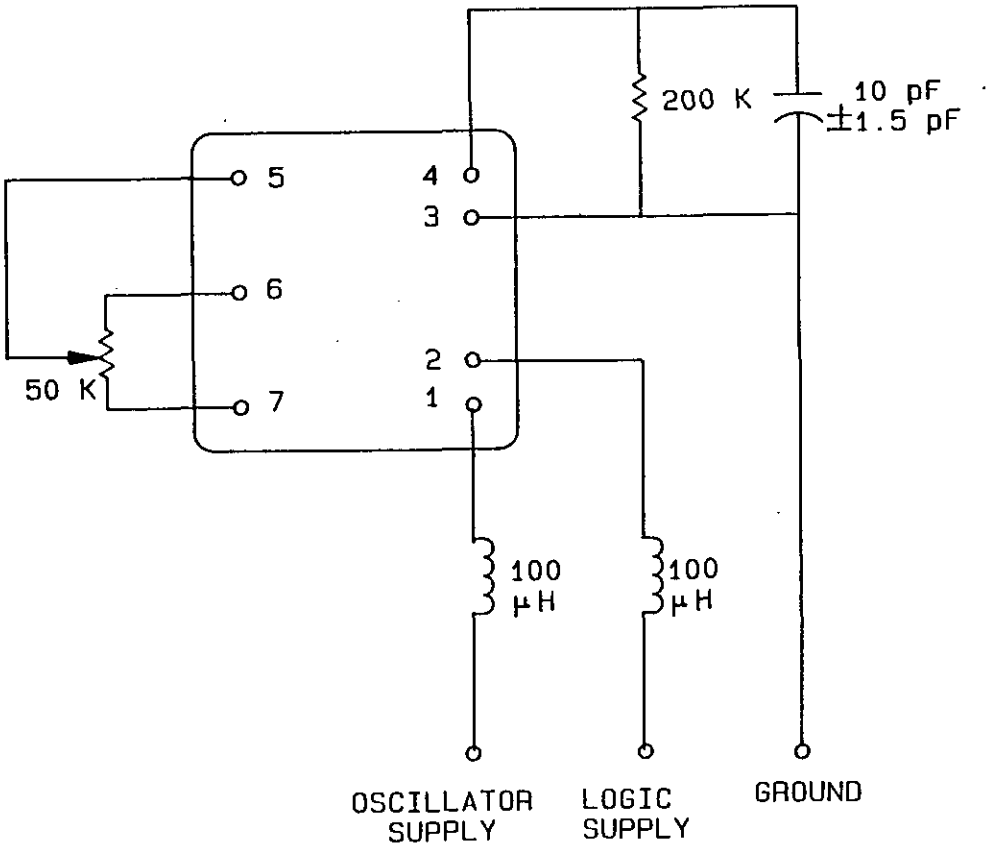
Initial frequency-temperature accuracy including hysteresis and trim effect: ±1.0 ppm (-55°C to +85°C and ±3.0 ppm trim). Measurements taken at 10°C increments.

Frequency-voltage tolerance: ±0.10 ppm for 10 V dc ±3%. Measurements taken at -55°C, +23°C, and +85°C.

Frequency-load tolerance: ±0.10 ppm for 10 pF ±1.5 pF (see figure 2).

Phase noise, steady state: See table I.

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NOTES
1. Component parts or equivalent.
2. When pins 5 and 6 are shorted, frequency shall increase.

FIGURE 2. Load test circuit.

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TABLE I. Phase noise.

Phase noise measured in a 1 hz bandwidth	
Offset from carrier	dBc
10 Hz	- 90
100 Hz	-120
1 kHz	-140
10 kHz	-140

Acceleration sensitivity, vibration (Group C inspection): When subjected to vibration (operating) specified herein, the acceleration sensitivity shall not exceed 1 x 10^{-8}/g.

Output waveform: Symmetrical square wave.

Output voltage (measured at -55°C, +23°C, and +85°C):

Logic 1: 0.9 V_{logic} minimum, V_{logic} + 0.5 V maximum.

Logic 0: -0.5 V minimum, 0.1 V_{logic} maximum.

Rise and fall time (measured at -55°C, +23°C, and +85°C): 20 nanoseconds maximum.

Duty cycle (measured at -55°C, +23°C, and +85°C): 40 minimum to 60 maximum percent.

Spurious response: At least -80 dB within 10 kHz to 100 MHz.

Startup time: 50 milliseconds maximum. Measurements taken at -55°C, +23°C, and +85°C.

Frequency aging: Frequency aging at 60°C shall not exceed the following limits:

Change between any successive days:	2.0 x 10^{-8}
Total change from day 0 to day 30:	2.0 x 10^{-7}
Projected change for 1.0 year:	6.0 x 10^{-7}

Conducted interference: 20 mV peak to peak (pins 1 and 2) and 200 mV peak to peak (pin 5) maximum within 10 Hz to 100 MHz (see figure 2). Measurement taken with minimum 1000 pF bypass capacitor inserted between pins 1, 2, and ground. Leads at pins 5, 6, and 7 shall not exceed 1.5 inches. Potentiometer shall be set approximately in the center and at the two extremes of its adjustment range.

Frequency-environment tolerance: ±0.30 ppm.

Vibration, sinusoidal (nonoperating): MIL-STD-202, method 204, test condition A, except upper frequency limit shall be 2000 Hz.

Vibration, sinusoidal (operating):

Frequency	Amplitude
10 Hz - 40 Hz	.06 in. DA ±10%
40 Hz - 2000 Hz	5 g (peak) ±10%

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Shock (nonoperating): MIL-STD-202, method 213, test condition I.

Thermal shock (nonoperating): MIL-STD-202, method 107, test condition B-1.

Ambient pressure (operating): MIL-STD-202, method 105, test condition B.

Storage temperature: In accordance with MIL-O-55310.

Resistance to soldering heat: MIL-STD-202, method 210, test condition E.

Moisture resistance: MIL-STD-202, method 106.

Salt spray (corrosion): MIL-STD-202, method 101, test condition B.

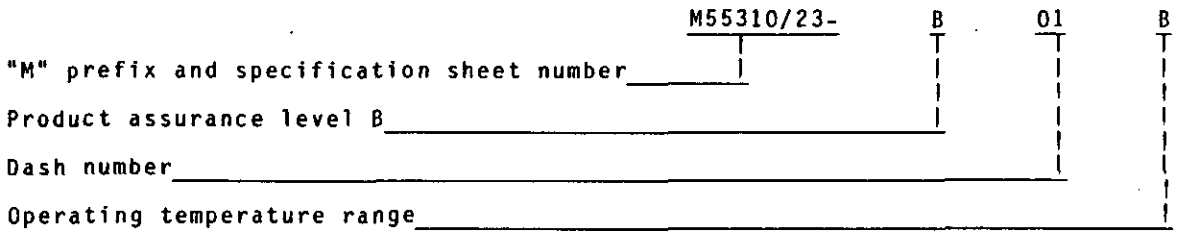
Terminal strength (lead integrity): MIL-STD-202, method 211, test condition C, applied force - 1 pound.

Solderability: MIL-STD-202, method 208.

Resistance to solvents: MIL-STD-202, method 215.

Screening: Post burn-in electrical tests - input current, output waveform, output voltage, startup time and initial frequency-temperature accuracy (half temperature cycle, condition I).

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash and coded alphas, and numeric number as shown in following example:



CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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